Experiment 3:

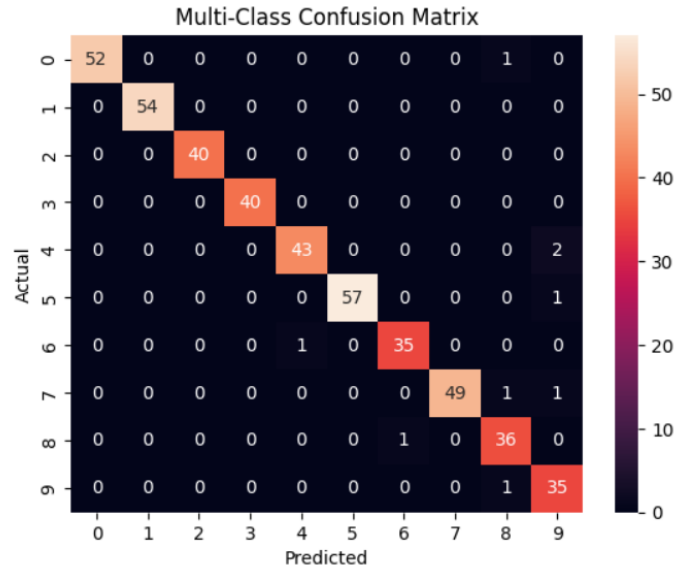
Aim: To analyse the performance of a multi-class confusion matrix using Python.

Code:

```
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns
X, y = load_digits(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25)
model = RandomForestClassifier()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Multi-Class Confusion Matrix")
plt.show()
```

Output:

```
Accuracy: 0.98
```



Result: Thus, the multi-class confusion matrix was successfully implemented using the Digits dataset and Random Forest classifier.